

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1529

Temple Pecan Creek 100 kV
September 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1529. The Interconnection Customer proposes a 109.0 MW solar plus storage facility interconnecting at the Temple Pecan Creek 100 kV substation in the Illinois area. The request was received on July 20, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 542 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$493,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
491 E 48Th Street - James St 100.0 kV	James St - Portuguese 100.0 kV Line	116.2	112	49.9
Jame St Substation - Portuguese 100.0 kV Line	James St Substation - 491 E 48Th St 100.0 kV Line	103.4	121	49.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
491 E 48Th Street - James St 100.0 kV	Rebuild line with higher-capacity conductor	\$1,050,000	34% (\$355,000)
James St - Portuguese 100.0 kV	Reconductor line	\$410,000	34% (\$138,000)
	Total allocated to project		\$493,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.